

ECE 253 – Digital Image Processing

Matrix-Based (Block) Transforms

Review - Complex vectors

Complex conjugate transpose

$$\mathbf{v}^H = (\mathbf{v}^*)^T = (\mathbf{v}^T)^*$$

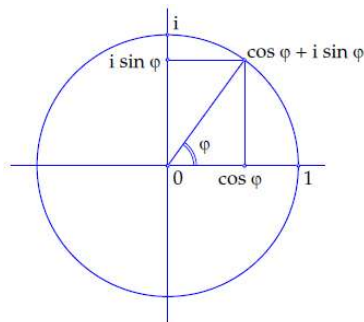
where the elements of $\mathbf{v}$ are complex numbers

$$C = R + jI, \text{ where } j = \sqrt{-1}$$

Real numbers are a subset of complex numbers where $I = 0$

Complex conjugate

$$C^* = R - jI$$



Unit Circle

Review - Complex vectors

Inner product

Real vectors

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v} = \sum_{i=0}^{N-1} u_i v_i = \mathbf{v}^T \mathbf{u} = \langle \mathbf{v}, \mathbf{u} \rangle$$

Complex vectors

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^H \mathbf{v} = (\mathbf{u}^*)^T \mathbf{v} = \sum_{i=0}^{N-1} u_i^* v_i = (\mathbf{v}^H \mathbf{u})^* = \langle \mathbf{v}, \mathbf{u} \rangle^*$$

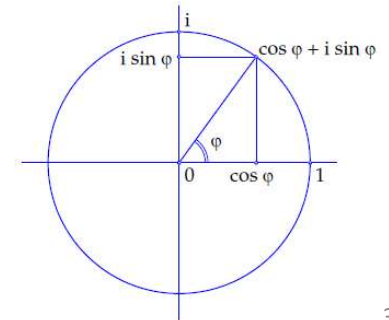
If the vectors are continuous functions on the interval $a \leq x \leq b$, then

$$\langle f(x), g(x) \rangle = \int_a^b f^*(x) g(x) dx$$

Complex multiplication

$$(a + jb)(c + jd) = (ac - db) + j(ad + bc)$$

$$(r_1 e^{jw_1})(r_2 e^{jw_2}) = (r_1 r_2 e^{j(w_1 + w_2)})$$



Review - Complex vectors

Properties

$$\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$$

$$\langle \mathbf{u}, \alpha \mathbf{v} \rangle = \alpha \langle \mathbf{u}, \mathbf{v} \rangle, \text{ where } \alpha \text{ is a scalar}$$

$$\langle \mathbf{v}, \mathbf{v} \rangle \geq 0, \text{ and } \langle \mathbf{v}, \mathbf{v} \rangle = 0 \text{ if and only if } \mathbf{v} = \mathbf{0}$$

The norm of a vector $\mathbf{z}$ is $\|\mathbf{z}\| = \sqrt{\langle \mathbf{z}, \mathbf{z} \rangle}$

If $\|\mathbf{z}\| = 1$, then $\mathbf{z}$ is called a unit vector

The angle between two nonzero vectors is

$$\theta = \cos^{-1} \frac{\langle \mathbf{z}, \mathbf{w} \rangle}{\|\mathbf{z}\| \|\mathbf{w}\|}$$

If $\theta = 90^\circ$, then $\mathbf{z}$ and $\mathbf{w}$ are called orthogonal (i.e., $\langle \mathbf{z}, \mathbf{w} \rangle = 0$)

Review - Basis vectors

A set of nonzero vectors $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2, \dots$ such that

$$\langle \mathbf{w}_k, \mathbf{w}_l \rangle = 0 \text{ for } k \neq l \quad (\text{mutually orthogonal})$$

are an **orthogonal** basis of the inner product space they are said to span. Further, if these basis vectors are unit vectors, then they are an **orthonormal** basis and

$$\langle \mathbf{w}_k, \mathbf{w}_l \rangle = \delta_{kl} = \begin{cases} 0 & \text{for } k \neq l \\ 1 & \text{for } k = l \end{cases}$$

Similarly, a set of vectors $\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2, \dots$ and a complementary set of dual vectors $\tilde{\mathbf{w}}_0, \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \dots$ such that

$$\langle \tilde{\mathbf{w}}_k, \mathbf{w}_l \rangle = 0 \text{ for } k \neq l \quad (\text{mutually biorthogonal})$$

are a **biorthogonal** basis. They are a **biorthonormal** basis if and only if

$$\langle \tilde{\mathbf{w}}_k, \mathbf{w}_l \rangle = \delta_{kl} = \begin{cases} 0 & \text{for } k \neq l \\ 1 & \text{for } k = l \end{cases}$$

Review - Basis vectors

Let $W = \{\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2, \dots\}$ be an orthogonal basis of inner product space V , and let $\mathbf{z} \in V$. Vector $\mathbf{z}$ can be expressed as a linear combination of basis vectors

$$\mathbf{z} = \alpha_0 \mathbf{w}_0 + \alpha_1 \mathbf{w}_1 + \alpha_2 \mathbf{w}_2 + \dots$$

The inner products of $\mathbf{z}$ and the i -th basis vector $\mathbf{w}_i$ is

$$\langle \mathbf{w}_i, \mathbf{z} \rangle = \langle \mathbf{w}_i, \alpha_0 \mathbf{w}_0 + \alpha_1 \mathbf{w}_1 + \alpha_2 \mathbf{w}_2 + \dots \rangle$$

$$\langle \mathbf{w}_i, \mathbf{z} \rangle = \alpha_0 \langle \mathbf{w}_i, \mathbf{w}_0 \rangle + \alpha_1 \langle \mathbf{w}_i, \mathbf{w}_1 \rangle + \dots + \alpha_i \langle \mathbf{w}_i, \mathbf{w}_i \rangle + \dots$$

$$\langle \mathbf{w}_i, \mathbf{z} \rangle = \alpha_i \langle \mathbf{w}_i, \mathbf{w}_i \rangle, \text{ solve for } \alpha_i$$

$$\frac{\langle \mathbf{w}_i, \mathbf{z} \rangle}{\langle \mathbf{w}_i, \mathbf{w}_i \rangle} = \alpha_i$$

If W is an **orthonormal** basis, then $\alpha_i = \langle \mathbf{w}_i, \mathbf{z} \rangle$ (because $\langle \mathbf{w}_i, \mathbf{w}_i \rangle = 1$)

Similarly, let $W = \{\mathbf{w}_0, \mathbf{w}_1, \mathbf{w}_2, \dots\}$ and $\tilde{W} = \{\tilde{\mathbf{w}}_0, \tilde{\mathbf{w}}_1, \tilde{\mathbf{w}}_2, \dots\}$ be a biorthogonal basis.

$$\langle \tilde{\mathbf{w}}_i, \mathbf{z} \rangle = \langle \tilde{\mathbf{w}}_i, \alpha_0 \mathbf{w}_0 + \alpha_1 \mathbf{w}_1 + \alpha_2 \mathbf{w}_2 + \dots \rangle$$

$$\langle \tilde{\mathbf{w}}_i, \mathbf{z} \rangle = \alpha_i \langle \tilde{\mathbf{w}}_i, \mathbf{w}_i \rangle, \text{ solve for } \alpha_i$$

$$\frac{\langle \tilde{\mathbf{w}}_i, \mathbf{z} \rangle}{\langle \tilde{\mathbf{w}}_i, \mathbf{w}_i \rangle} = \alpha_i$$

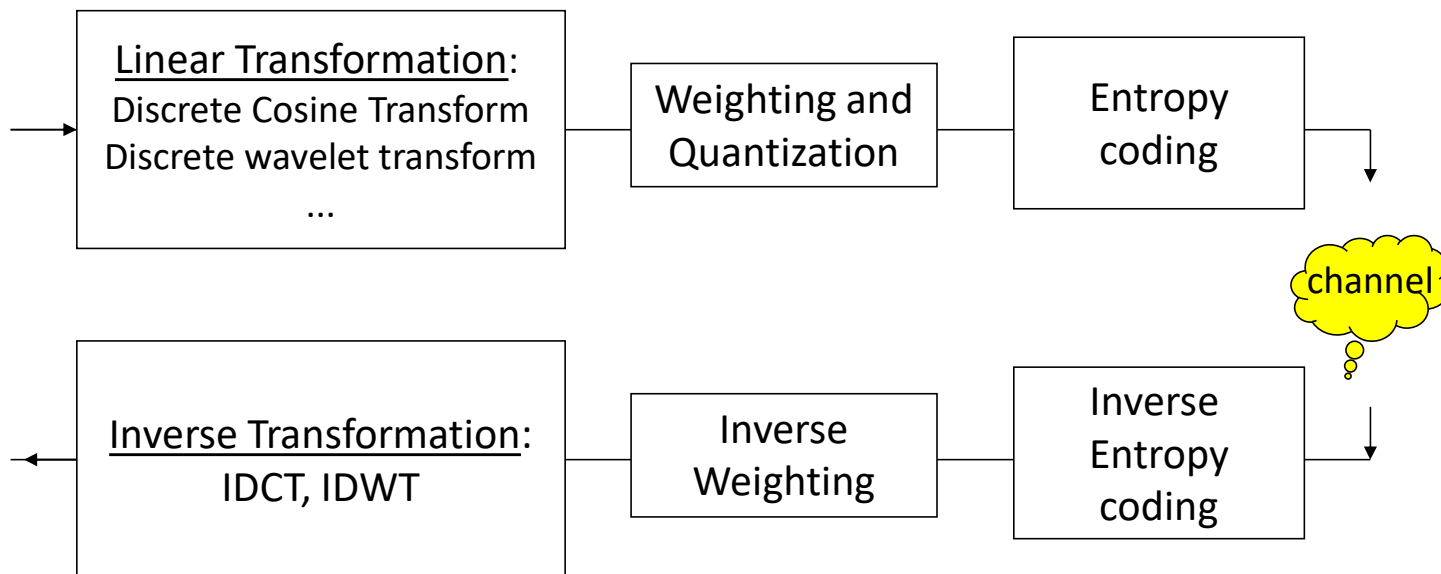
If W and $\tilde{W}$ are a **biorthonormal** basis, then $\alpha_i = \langle \tilde{\mathbf{w}}_i, \mathbf{z} \rangle$ (because $\langle \tilde{\mathbf{w}}_i, \mathbf{w}_i \rangle = 1$)

Matrix-based (Block) transforms

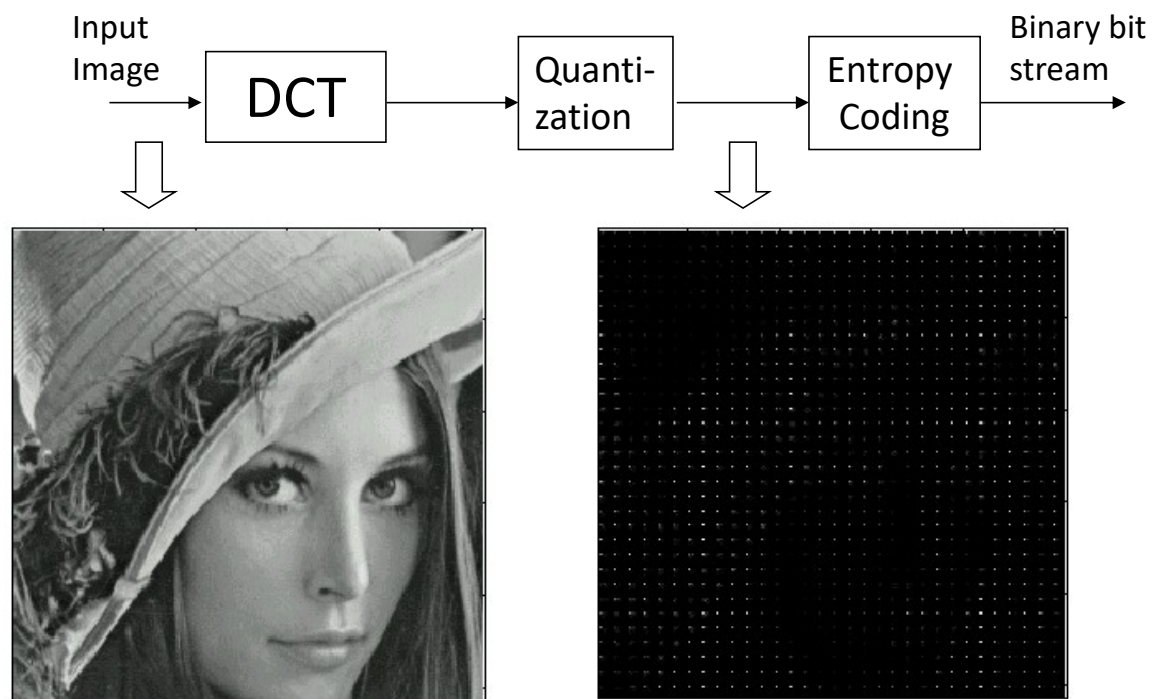




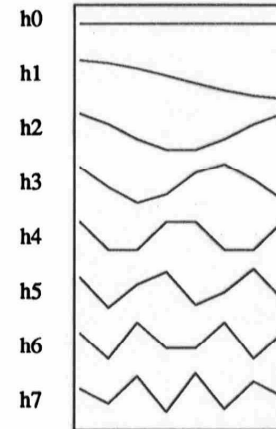
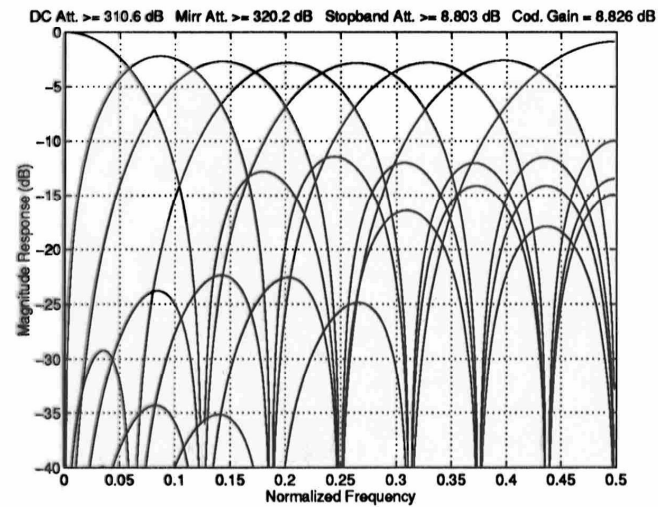
Transform-Based Image Coding



JPEG

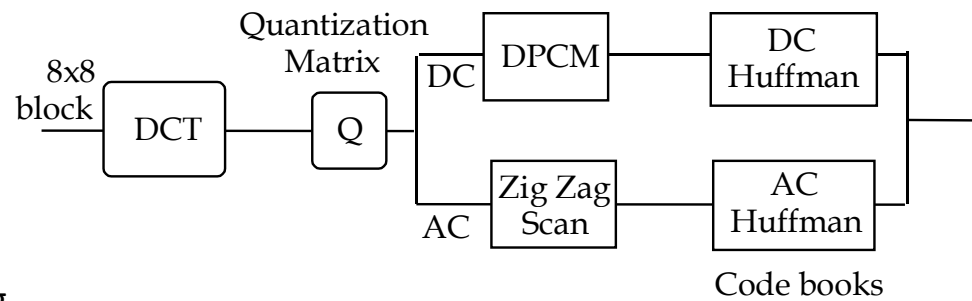


DCT in Frequency and Time Domain

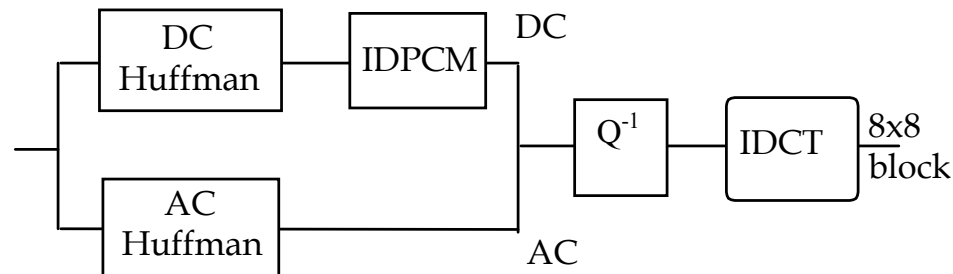


JPEG Encoding and Decoding

Encoding

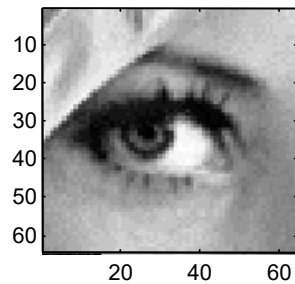


Decoding

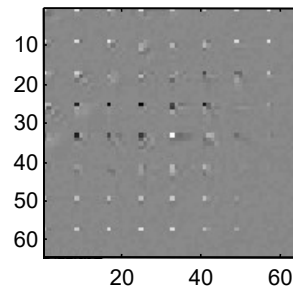


www.jpeg.org

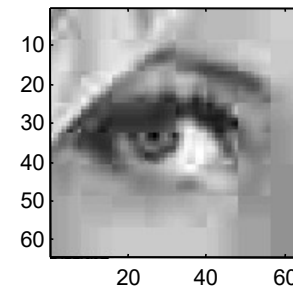
JPEG Coding Example



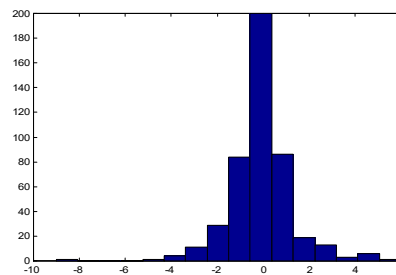
Original 64x64 image at 8 bits/pixel (8 bpp). Total $64 \times 64 \times 8 = 32768$ bits. After encoding, 287 bits for DC coefficient, 1364 bits for AC coefficients. Compression ratio: 19.85:1 or 0.403bpp.



Block DCT coefficients



Decoded image (19.85:1 ratio) 0.403 bpp. Note the blocking effect.



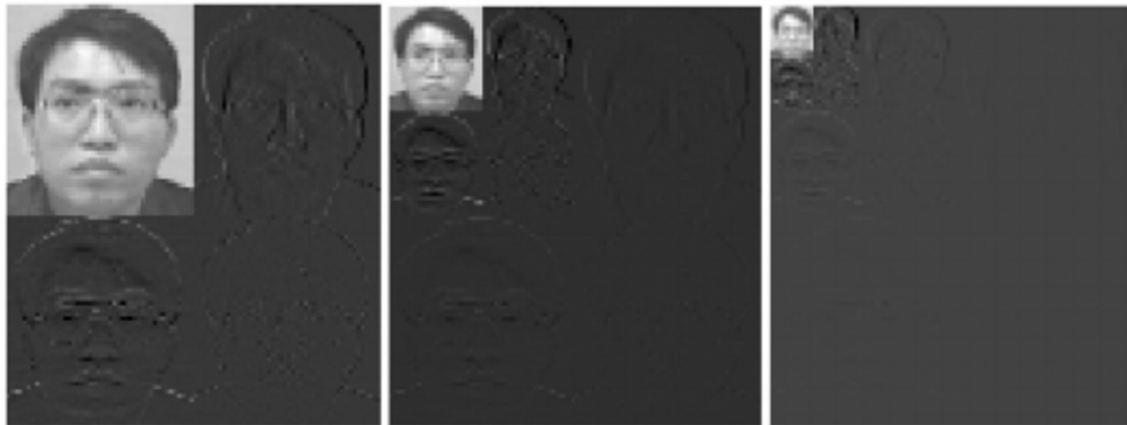
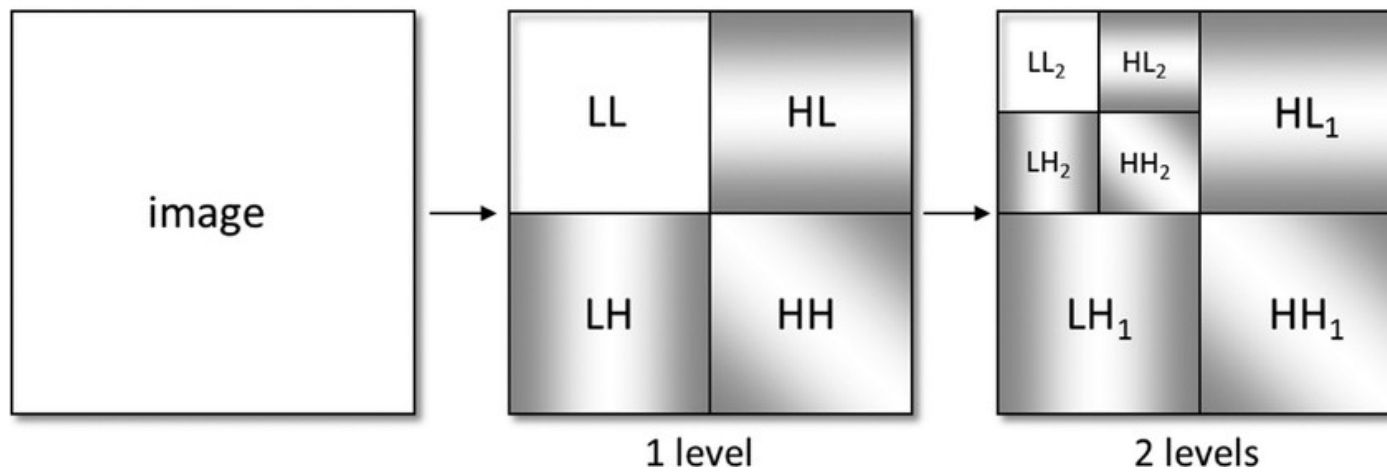
Histogram of AC coefficients

Disadvantages of DCT

- DCT based JPEG uses blocks of image, there is still correlation across blocks.
- Block boundaries are noticeable in some cases
- Blocking artifacts at low bit rates
- Can overlap the blocks → Computationally expensive

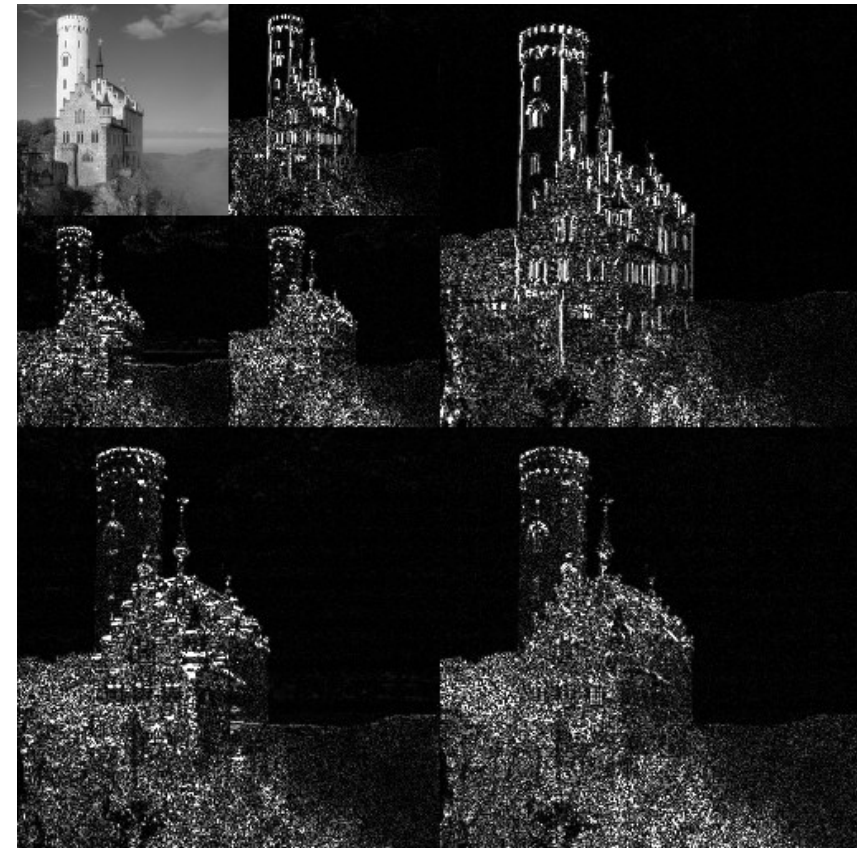
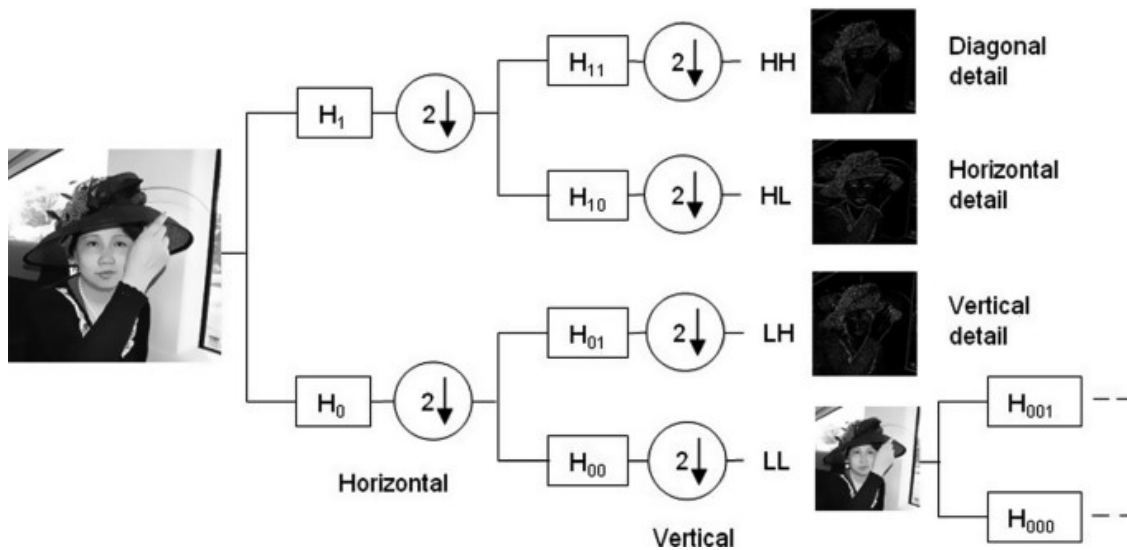


Multiresolution Representation



Courtesy of Boualem Boashash & Jen-Tzung Chien

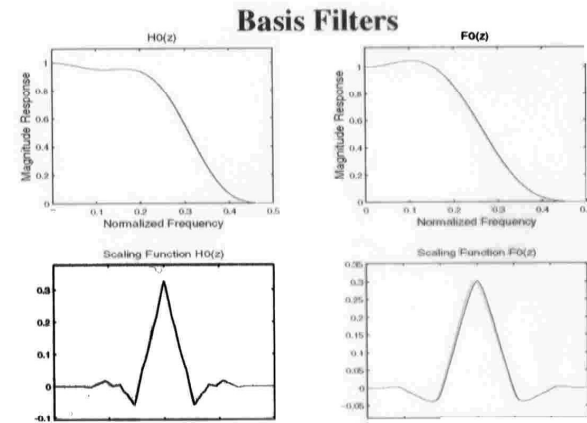
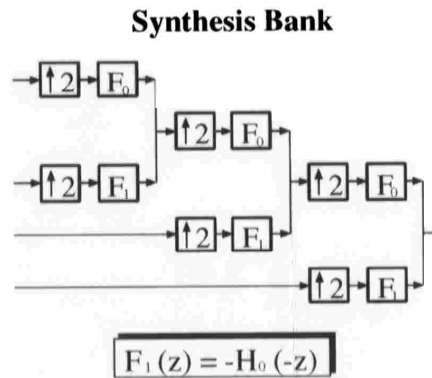
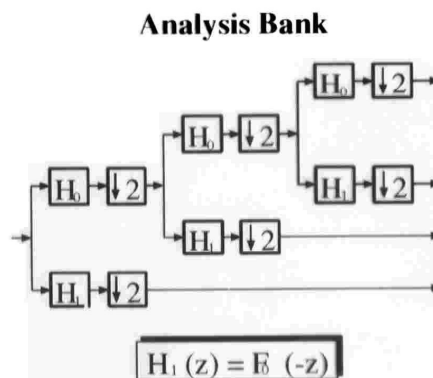
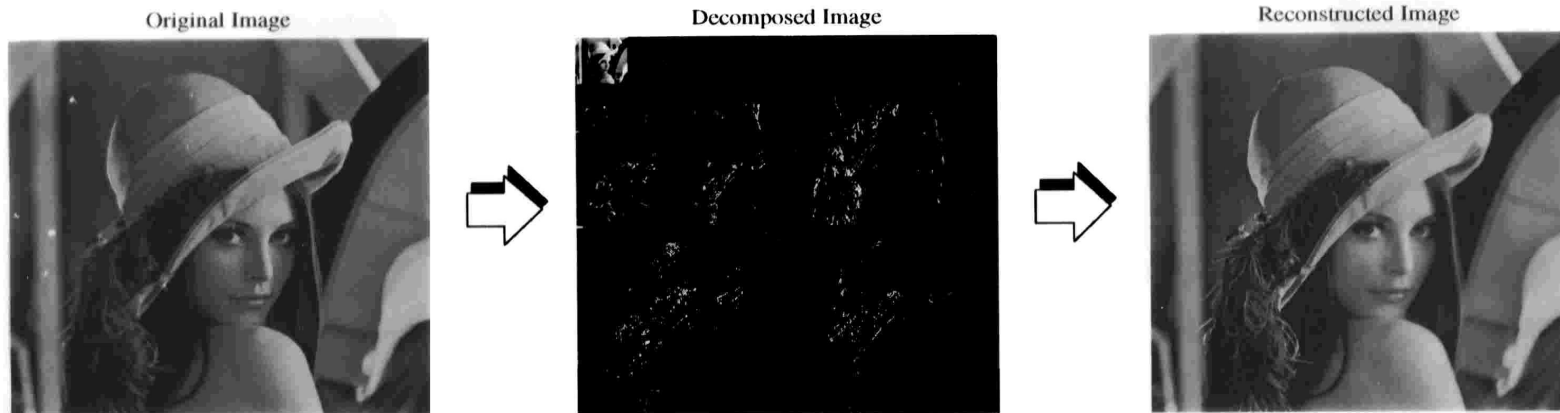
Multiresolution Representation



Why Wavelets?

- No need to block the image
- More robust under transmission errors
- Facilitates progressive transmission of the image (Scalability)

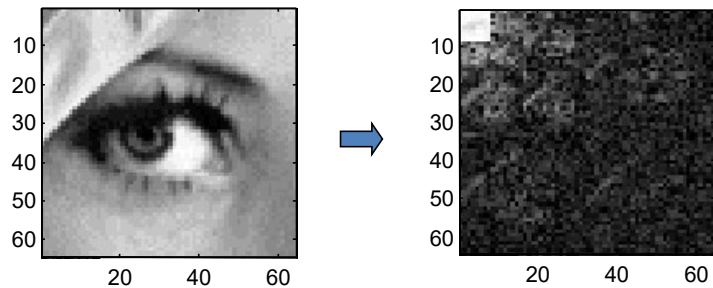
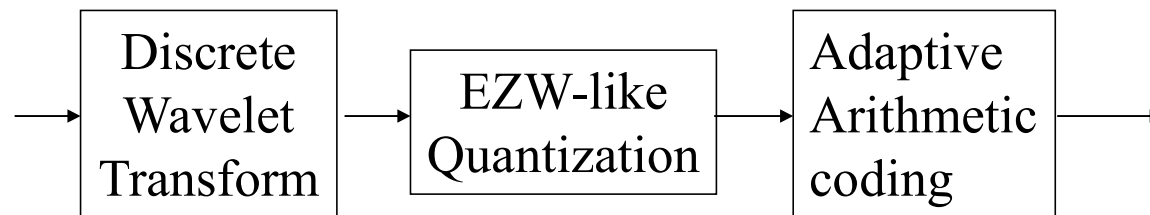
WAVELET TRANSFORM



Important Issues:

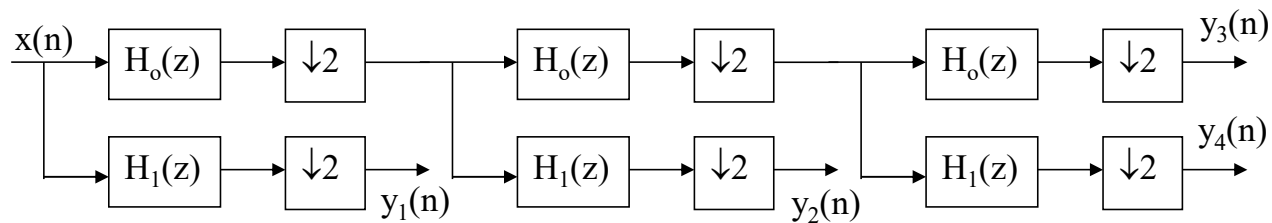
- linear phase (symmetric)
- perfect reconstruction
- smoothness of the basis functions
- vanishing moments
- DC leakage, mirror frequencies
- nonuniform filter bank

Wavelet Based Coding

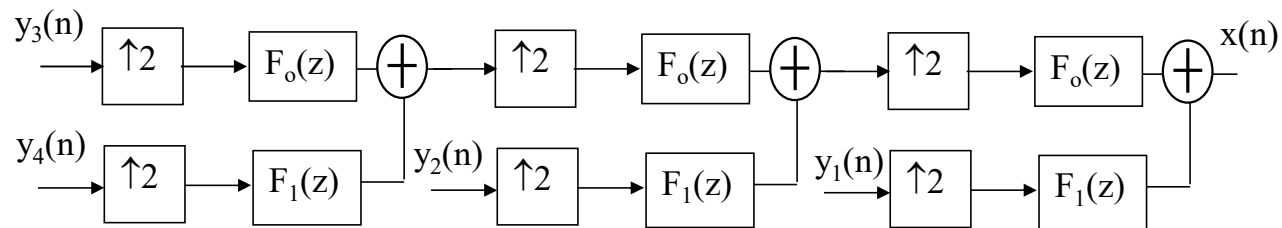


2D discrete wavelet transform converts images into “sub-bands”
Upper left is the DC coefficient
Lower right are higher frequency sub-bands.

Discrete Wavelet Transform

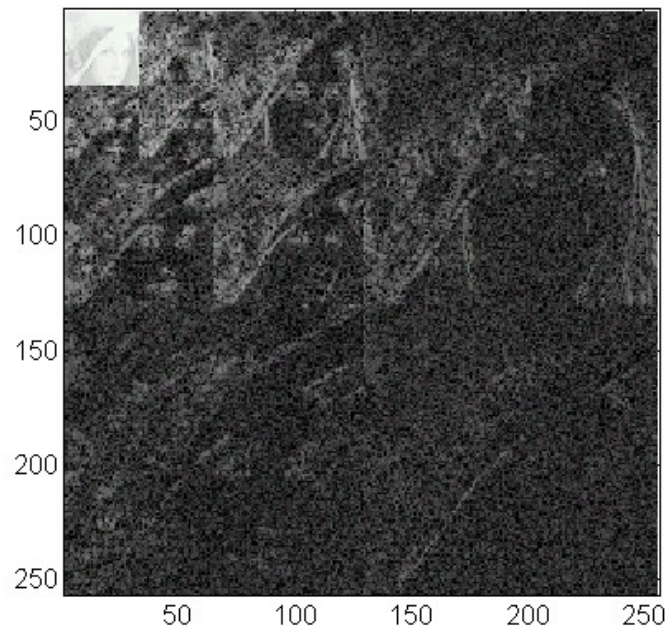


1D Analysis Filter



1D Synthesis Filter

Wavelet Coefficients



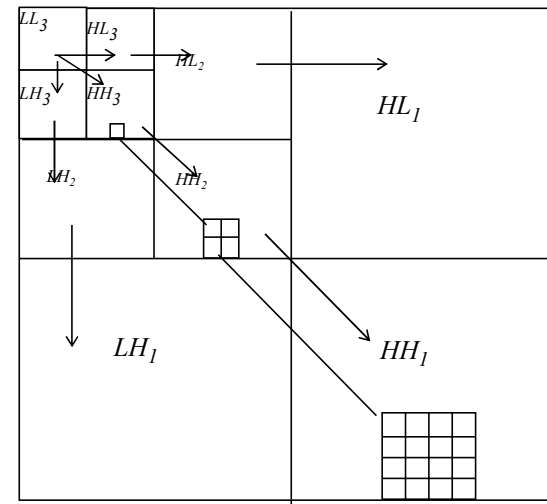
Wavelet transform coefficients are arranged so that low frequency sub-band is at upper left corner.

There is correlation among sub-bands of different scales corresponding to same location!

DWT for Image Compression

- **Image Decomposition**

- Parent
- Children
- Descendants:
corresponding coeff. at
finer scales
- Ancestors: corresponding
coeff. at coarser scales



–Parent-children dependencies of subbands: arrow points from the subband of parents to the subband of children.

DWT for Image Compression

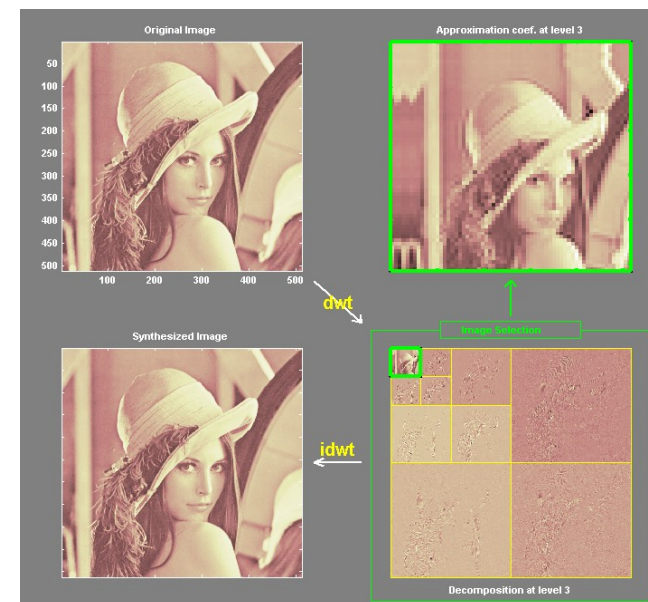
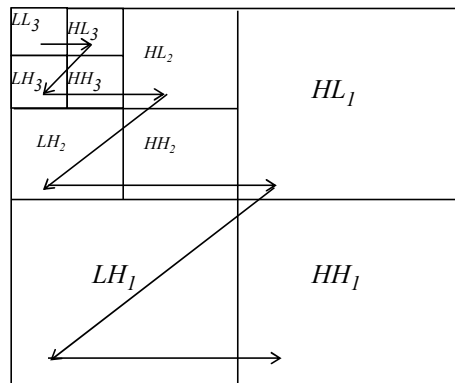
- **Image Decomposition**

- Feature 1:

- Energy distribution concentrated in low frequencies

- Feature 2:

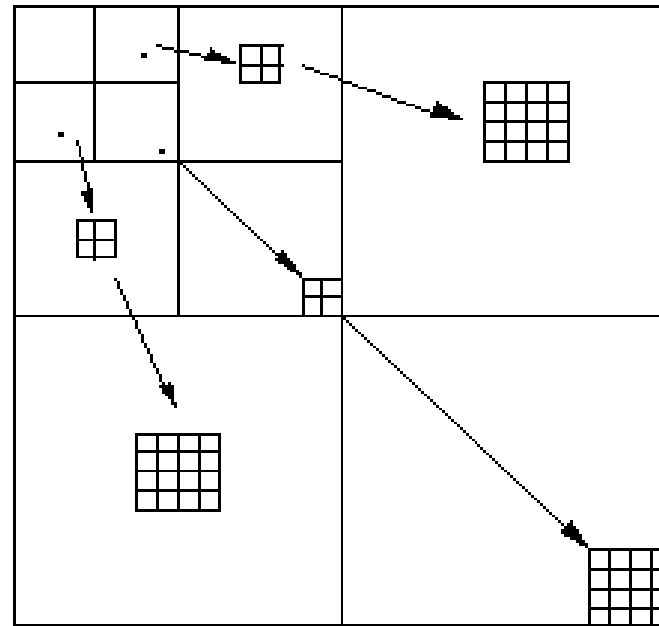
- Spatial self-similarity across subbands



The scanning order of the subbands for encoding the significance map.

Embedded Zerotree Wavelet Compression (EZW)

- The zerotree is based on the hypothesis that if a wavelet coefficient at a coarse scale is insignificant, then all wavelet coefficients of the same orientation in the same spatial location at finer scales are likely to be insignificant.
 - Natural images in general have a low pass spectrum. When an image is wavelet transformed, the energy in the sub-bands decreases with the scale goes higher so the wavelet coefficient will, on average, be smaller in the higher levels.



Scalability



Decompressed image "bike" at (a) 0.125 b/p, (b) 0.25 b/p, (c) 0.5 b/p



Decode at different resolutions

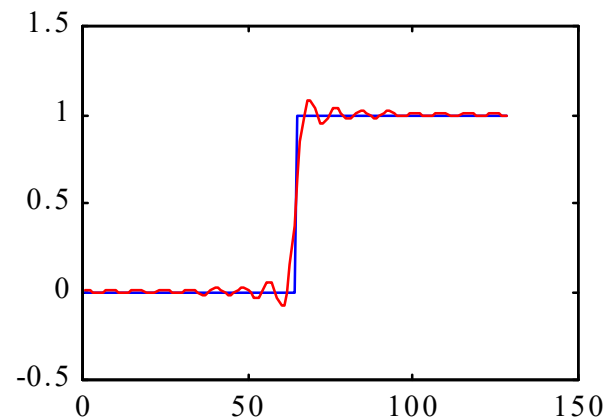
ringing Artifact



Ringing -- Spurious ripples along major edges. Visible with bright, smooth background.

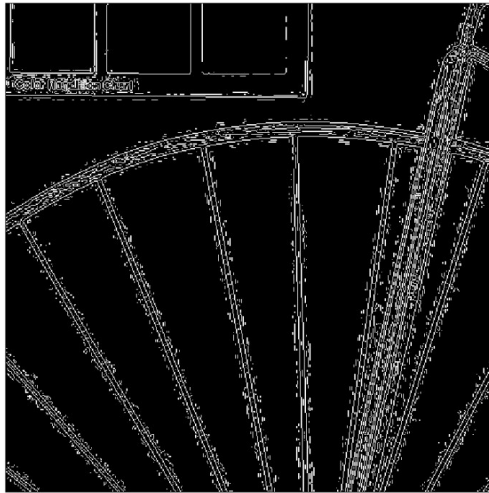
Cause of Ringing Artifact

- Ringing artifact is a result of Gibbs phenomenon when long basis functions are used for image transformation.
- Gibbs phenomenon -- Truncation of frequency domain coefficients leads to oscillations in spatial domain around step discontinuities.

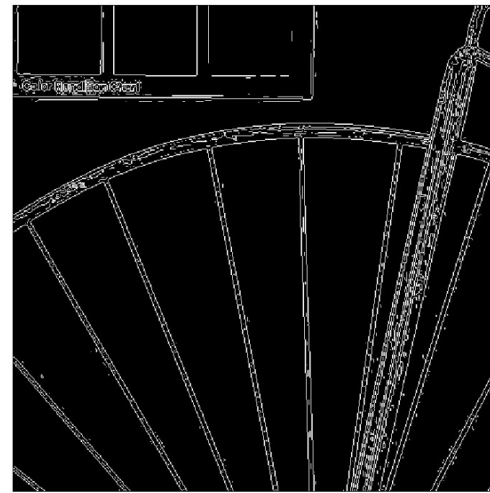


Truncation of DCT coefficients from 128 down to 32 leads to ringing artifact.

Elimination of Ringing (false edges)

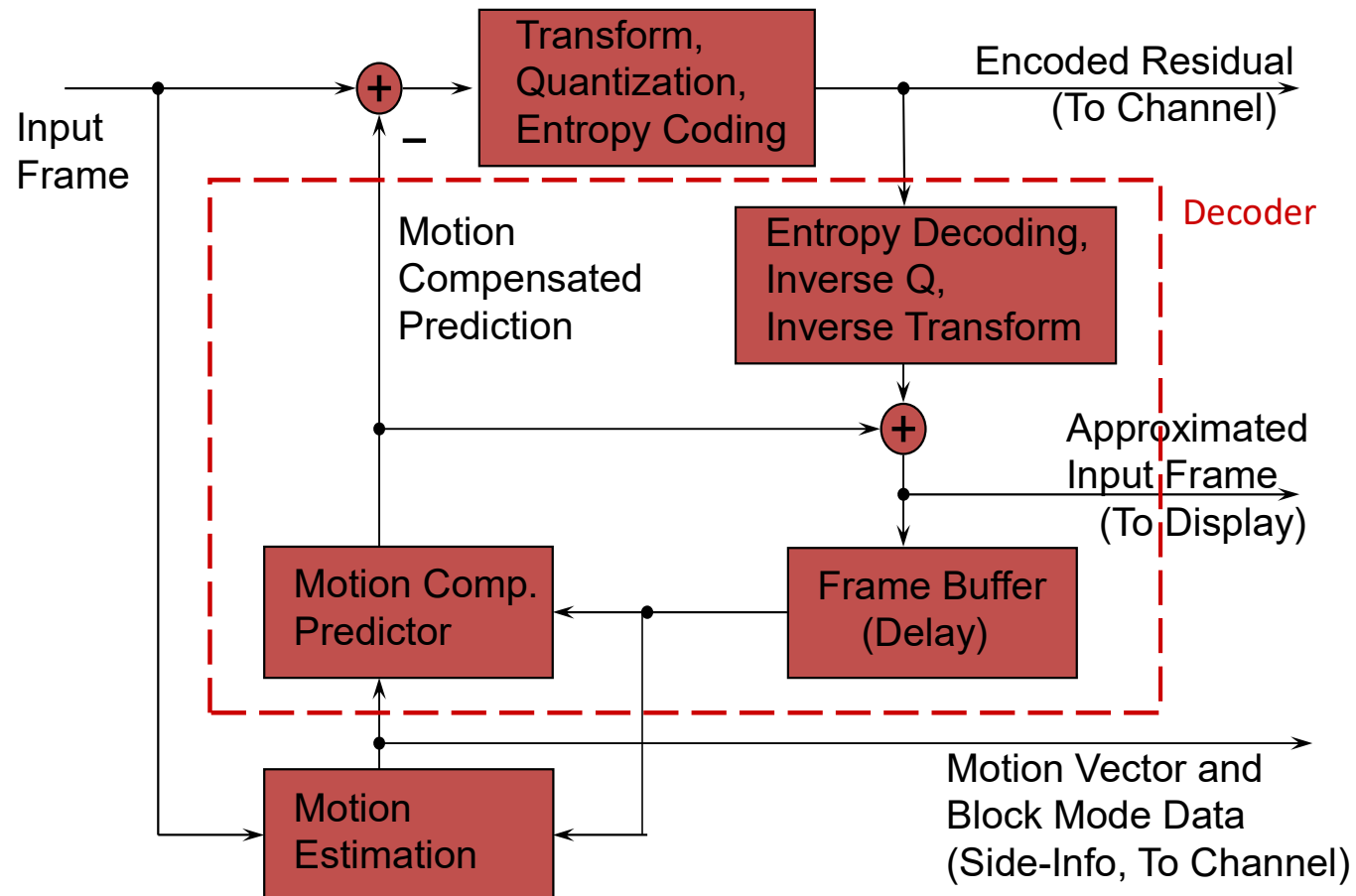


JPEG2000

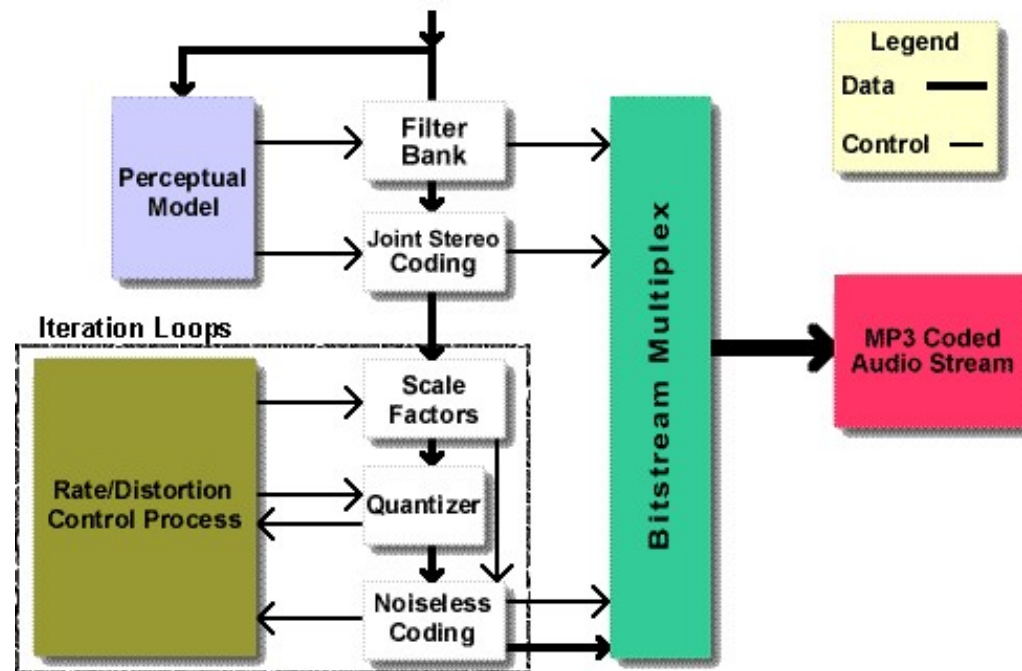


Proposed Method

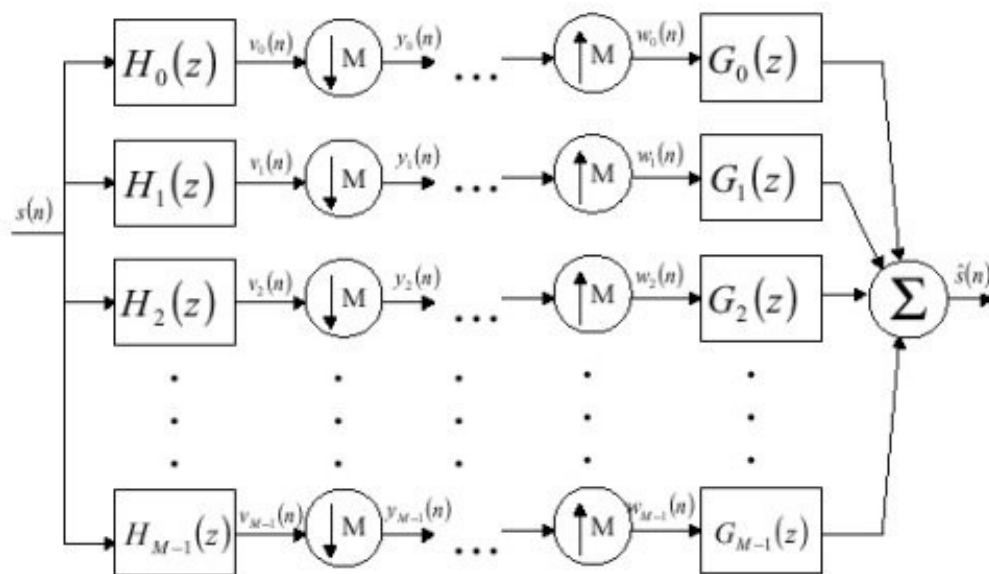
Typical Motion-Compensation Codec (MPEG2, H.264, SVC)



MP3 Audio Codec

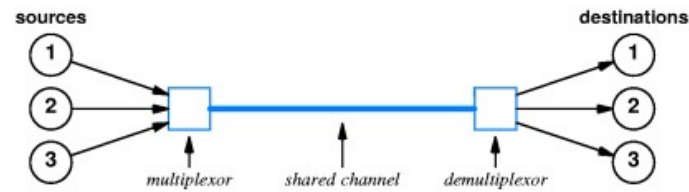


Filter Banks



Discrete Cosine Transform
 Fourier Transform, FFT
 Short-time Fourier Transform

Frequency Division Multiplexing (FDM)



Matrix-based (block) transforms

1D forward transform

$$T(u) = \sum_{x=0}^{N-1} f(x)r(x, u)$$

$T(u)$ is the transform of $f(x)$

x is a spatial variable

u is a transform variable

$r(x, u)$ is a forward transformation kernel

1D inverse transform

$$f(x) = \sum_{u=0}^{N-1} T(u)s(x, u)$$

$f(x)$ is the inverse transform of $T(u)$

u is a transform variable

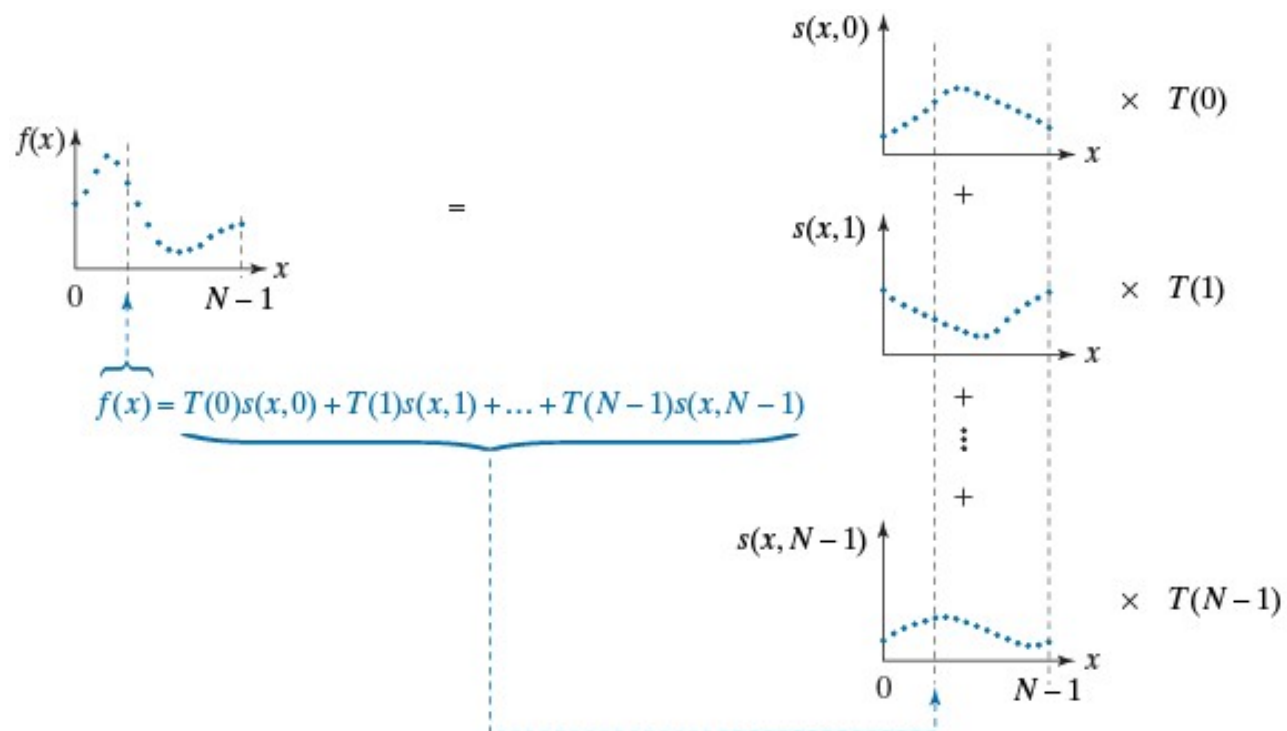
x is a spatial variable

$s(x, u)$ is an inverse transformation kernel

Note the transformation kernels $r(x, u)$ and $s(x, u)$ depend only on x and u , not values of $f(x)$ and $T(u)$

General inverse transform using basis vectors

$$f(x) = \sum_{u=0}^{N-1} T(u)s(x, u)$$



Matrix-based transforms - orthonormal basis vectors

$$\langle s(x, u), f(x) \rangle = T(0)\langle s(x, u), s(x, 0) \rangle + T(1)\langle s(x, u), s(x, 1) \rangle + \cdots + T(u)\langle s(x, u), s(x, u) \rangle + \cdots$$

$$\langle s(x, u), f(x) \rangle = T(u)\langle s(x, u), s(x, u) \rangle$$

$$\frac{\langle s(x, u), f(x) \rangle}{\langle s(x, u), s(x, u) \rangle} = T(u)$$

If $s(x, 0), s(x, 1), s(x, 2), \dots$ are orthonormal basis vectors, then

$$\langle s(x, u), f(x) \rangle = T(u)$$

In vector form

$$T(u) = \langle s(x, u), f(x) \rangle$$

$$T(u) = \langle \mathbf{s}_u, \mathbf{f} \rangle$$

$$\text{where } \mathbf{s}_u = \begin{matrix} x \downarrow \\ \begin{bmatrix} s(0, u) \\ s(1, u) \\ \vdots \\ s(N-1, u) \end{bmatrix} \end{matrix} \quad \text{and } \mathbf{f} = \begin{matrix} x \downarrow \\ \begin{bmatrix} f(0) \\ f(1) \\ \vdots \\ f(N-1) \end{bmatrix} \end{matrix}$$

$$T(u) = \mathbf{s}_u^H \mathbf{f} \text{ for complex vectors}$$

$$T(u) = \mathbf{s}_u^T \mathbf{f} \text{ for real vectors}$$

Matrix-based transforms - orthonormal basis vectors

In matrix form

$$\begin{array}{c}
 \begin{array}{c} \downarrow u \\ \left[\begin{array}{c} T(0) \\ T(1) \\ \vdots \\ T(N-1) \end{array} \right] \end{array} \begin{array}{c} \xrightarrow{x} \\ u \end{array} = \begin{array}{c} \left[\begin{array}{c} \mathbf{s}_0^H \\ \mathbf{s}_1^H \\ \vdots \\ \mathbf{s}_{N-1}^H \end{array} \right] \end{array} \begin{array}{c} \left[\begin{array}{c} f(0) \\ f(1) \\ \vdots \\ f(N-1) \end{array} \right] \end{array} \quad \text{for complex vectors} \\
 \\
 \begin{array}{c} \left[\begin{array}{c} T(0) \\ T(1) \\ \vdots \\ T(N-1) \end{array} \right] \end{array} = \begin{array}{c} \left[\begin{array}{c} \mathbf{s}_0^T \\ \mathbf{s}_1^T \\ \vdots \\ \mathbf{s}_{N-1}^T \end{array} \right] \end{array} \begin{array}{c} \left[\begin{array}{c} f(0) \\ f(1) \\ \vdots \\ f(N-1) \end{array} \right] \end{array} \quad \text{for real vectors} \\
 \\
 \mathbf{t} = \mathbf{A}\mathbf{f}
 \end{array}$$

where transformation matrix

$$\begin{array}{c}
 \begin{array}{c} \xrightarrow{x} \\ u \end{array} \\
 \mathbf{A} = \begin{array}{c} \left[\begin{array}{c} \mathbf{s}_0^H \\ \mathbf{s}_1^H \\ \vdots \\ \mathbf{s}_{N-1}^H \end{array} \right] \end{array} \quad \begin{array}{c} N \times N \\ \text{for complex vectors} \end{array} \\
 \\
 \mathbf{A} = \begin{array}{c} \left[\begin{array}{c} \mathbf{s}_0^T \\ \mathbf{s}_1^T \\ \vdots \\ \mathbf{s}_{N-1}^T \end{array} \right] \end{array} \quad \begin{array}{c} N \times N \\ \text{for real vectors} \end{array}
 \end{array}$$

Matrix-based transforms - orthonormal basis vectors

Forward and inverse transformations

$$\mathbf{t} = \mathbf{A}\mathbf{f}$$

$$\mathbf{A}^{-1}\mathbf{t} = \mathbf{f}$$

$$\mathbf{A}^H\mathbf{t} = \mathbf{f} \text{ for complex } \mathbf{A}$$

$$\mathbf{A}^T\mathbf{t} = \mathbf{f} \text{ for real } \mathbf{A}$$

Note forward transformation kernel $r(x, u)$ is not needed

Properties

Unitary matrix $\mathbf{A}$

$$\mathbf{A}^H\mathbf{A} = \mathbf{A}\mathbf{A}^H = \mathbf{A}^*\mathbf{A}^T = \mathbf{A}^T\mathbf{A}^* = \mathbf{I}$$

$$\mathbf{A}^{-1} = \mathbf{A}^H$$

Orthogonal matrix $\mathbf{A}$

$$\mathbf{A}^T\mathbf{A} = \mathbf{A}\mathbf{A}^T = \mathbf{I}$$

$$\mathbf{A}^{-1} = \mathbf{A}^T$$

Matrix-based transforms – Biorthonormal vectors

In matrix form

$$\begin{bmatrix} \downarrow u & T(0) \\ & T(1) \\ & \vdots \\ & T(N-1) \end{bmatrix} = \begin{bmatrix} \downarrow u & \tilde{s}_0^H \\ & \tilde{s}_1^H \\ & \vdots \\ & \tilde{s}_{N-1}^H \end{bmatrix} \begin{bmatrix} f(0) \\ f(1) \\ \vdots \\ f(N-1) \end{bmatrix} \quad \text{for complex vectors}$$

$$\begin{bmatrix} T(0) \\ T(1) \\ \vdots \\ T(N-1) \end{bmatrix} = \begin{bmatrix} \tilde{s}_0^T \\ \tilde{s}_1^T \\ \vdots \\ \tilde{s}_{N-1}^T \end{bmatrix} \begin{bmatrix} f(0) \\ f(1) \\ \vdots \\ f(N-1) \end{bmatrix} \quad \text{for real vectors}$$

$$\mathbf{t} = \tilde{\mathbf{A}}\mathbf{f}$$

$$\mathbf{f} = \mathbf{A}^H \mathbf{t} \quad \text{for complex vectors}$$

$$\mathbf{f} = \mathbf{A}^T \mathbf{t} \quad \text{for real vectors}$$

Use dual vectors for forward transformation only

Matrix-based transform

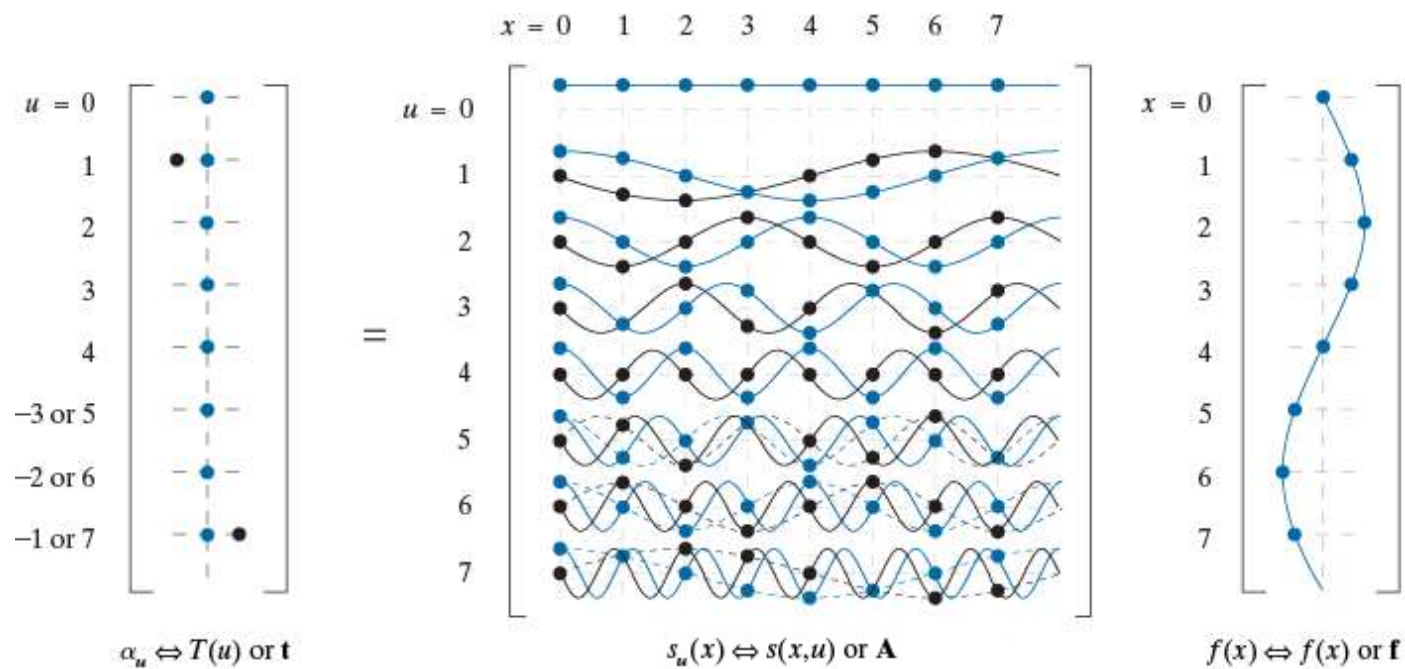
Example: 1D discrete Fourier transform (DFT)

$$T(u) = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} f(x) e^{-j2\pi ux/N} \text{ for } u = 0, 1, 2, \dots, N-1$$

$$\mathbf{t} = \mathbf{A}\mathbf{f}, \text{ where } \mathbf{A} = \begin{bmatrix} \mathbf{s}_0^H \\ \mathbf{s}_1^H \\ \vdots \\ \mathbf{s}_{N-1}^H \end{bmatrix}, \text{ where } s_u(x) = \frac{1}{\sqrt{N}} e^{j2\pi ux/N}$$

Matrix-based transform

Example: $N = 8$, DFT of $f(x) = \sin(2\pi x/N)$



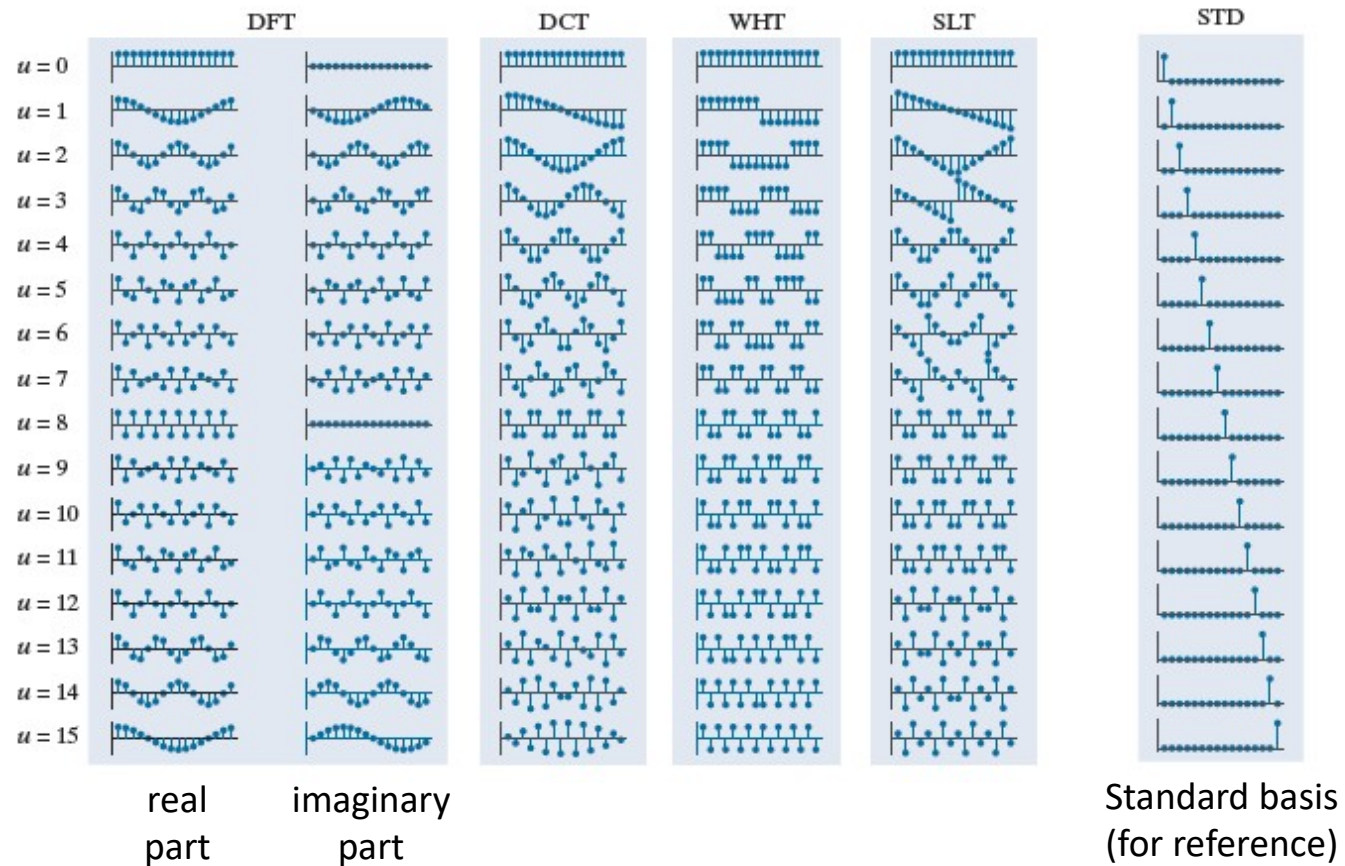
real part + imaginary part

Matrix-based transforms

- Discrete Fourier transform (DFT)
- Discrete Hartley transform (DHT)
- Discrete cosine transform (DCT)
- Discrete sine transform (DST)
- Walsh-Hadamard (WHT)
- Slant (SLT)
- Haar (HAAR)
- Daubechies (DB4)
- Biorthogonal B-spline (BIOR3.1)

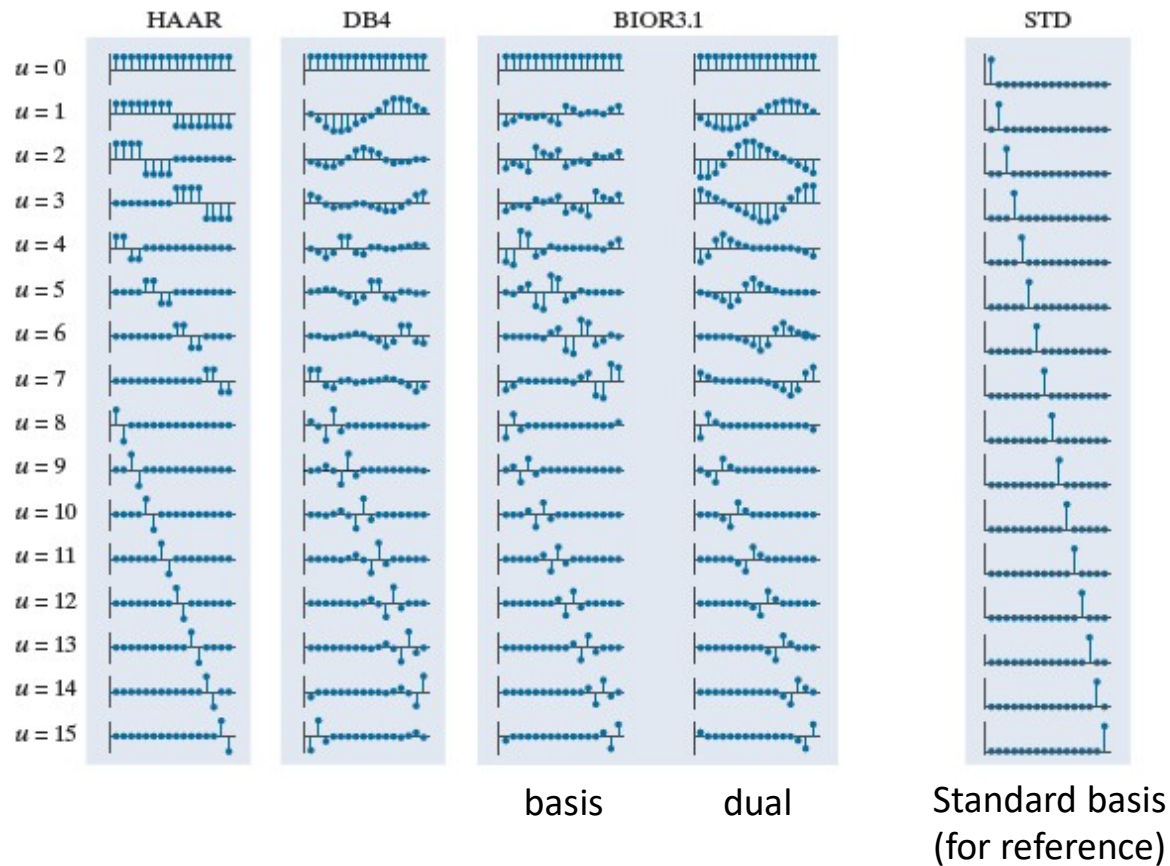
Basis vectors of matrix-based 1D transforms

$N = 16$



Basis vectors of matrix-based 1D transforms

$N = 16$



Matrix-based transforms in two dimensions

$$T(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) r(x, y, u, v) \quad \text{Forward transform}$$

$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} T(u, v) s(x, y, u, v) \quad \text{Inverse transform}$$

where

x, y are spatial variables

u, v are transform variables

$T(u, v)$ is the transform of $f(x, y)$

$f(x, y)$ is the inverse transform of $T(u, v)$

$r(x, y, u, v)$ is a forward transformation kernel

$s(x, y, u, v)$ is an inverse transformation kernel

Matrix-based transforms in two dimensions

If $s(x, y, u, v) = s_1(x, u)s_2(y, v)$, then s is called separable.

If $s(x, y, u, v) = s_1(x, u)s_1(y, v)$, then s is called symmetric.

If r and s are separable and symmetric, then

$$\mathbf{T} = \mathbf{A}_M \mathbf{F} \mathbf{A}_N^T$$

$$\mathbf{F} = \mathbf{A}_M^{*T} \mathbf{T} \mathbf{A}_N^*$$
 for complex vectors

$$\mathbf{F} = \mathbf{A}_M^T \mathbf{T} \mathbf{A}_N$$
 for real vectors

Note forward
transformation
kernel is not needed

where

$$\mathbf{T} = \begin{matrix} & \begin{matrix} \xrightarrow{v} \\ \downarrow u \end{matrix} & \begin{bmatrix} T(0,0) & T(0,1) & \cdots & T(0,N-1) \\ T(1,0) & T(1,1) & \cdots & T(1,N-1) \\ \vdots & \vdots & \ddots & \vdots \\ T(M-1,0) & T(M-1,1) & \cdots & T(M-1,N-1) \end{bmatrix} & , & M \times N \end{matrix}$$

$$\mathbf{A}_M = \begin{matrix} & \begin{matrix} \xrightarrow{x} \\ \downarrow u \end{matrix} & \begin{bmatrix} s_1(0,0) & s_1(1,0) & \cdots & s_1(M-1,0) \\ s_1(0,1) & s_1(1,1) & \cdots & s_1(M-1,1) \\ \vdots & \vdots & \ddots & \vdots \\ s_1(0,M-1) & s_1(1,M-1) & \cdots & s_1(M-1,M-1) \end{bmatrix} & , \text{ and } & M \times M \end{matrix}$$

$$\mathbf{A}_N = \begin{matrix} & \begin{matrix} \xrightarrow{y} \\ \downarrow v \end{matrix} & \begin{bmatrix} s_1(0,0) & s_1(1,0) & \cdots & s_1(N-1,0) \\ s_1(0,1) & s_1(1,1) & \cdots & s_1(N-1,1) \\ \vdots & \vdots & \ddots & \vdots \\ s_1(0,N-1) & s_1(1,N-1) & \cdots & s_1(N-1,N-1) \end{bmatrix} & & N \times N \end{matrix}$$

Transformation
matrices

Matrix-based transforms in two dimensions

If r and s are separable and symmetric, and $M = N$, then

- For **orthonormal** basis vectors

$$T = AFA^T$$

$$F = A^{*T}TA^* \text{ for complex vectors}$$

$$F = A^T TA \text{ for real vectors}$$

- For **biorthonormal** basis vectors

$$T = \tilde{A}F\tilde{A}^T$$

$$F = A^{*T}TA^* \text{ for complex vectors}$$

$$F = A^T TA \text{ for real vectors}$$

Matrix-based transforms in two dimensions

Forward transform

$$T(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) r(x, y, u, v)$$

where

$$T(u, v) = \begin{bmatrix} \overset{v}{\downarrow} T(0, 0) & T(0, 1) & \cdots & T(0, N-1) \\ T(1, 0) & T(1, 1) & \cdots & T(1, N-1) \\ \vdots & \vdots & \ddots & \vdots \\ \overset{u}{\downarrow} T(N-1, 0) & T(N-1, 1) & \cdots & T(N-1, N-1) \end{bmatrix}$$

$$r(x, y, u, v) = \begin{bmatrix} r(x, y, 0, 0) & r(x, y, 0, 1) & \cdots & r(x, y, 0, N-1) \\ r(x, y, 1, 0) & r(x, y, 1, 1) & \cdots & r(x, y, 1, N-1) \\ \vdots & \vdots & \ddots & \vdots \\ r(x, y, N-1, 0) & r(x, y, N-1, 1) & \cdots & r(x, y, N-1, N-1) \end{bmatrix}$$

Matrix-based transforms in two dimensions

Inverse transform

$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} T(u, v) s(x, y, u, v) \quad \text{Using transformation kernel}$$

$$F = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} T(u, v) S_{u,v} \quad \text{Using basis images}$$

where

$$F = \begin{matrix} \begin{matrix} x \downarrow \\ y \rightarrow \end{matrix} & \begin{bmatrix} f(0, 0) & f(0, 1) & \cdots & f(0, N-1) \\ f(1, 0) & f(1, 1) & \cdots & f(1, N-1) \\ \vdots & \vdots & \ddots & \vdots \\ f(N-1, 0) & f(N-1, 1) & \cdots & f(N-1, N-1) \end{bmatrix} \end{matrix}$$

Each $S_{u,v}$ is a basis image

$$S_{u,v} = \begin{matrix} \begin{matrix} x \downarrow \\ y \rightarrow \end{matrix} & \begin{bmatrix} s(0, 0, u, v) & s(0, 1, u, v) & \cdots & s(0, N-1, u, v) \\ s(1, 0, u, v) & s(1, 1, u, v) & \cdots & s(1, N-1, u, v) \\ \vdots & \vdots & \ddots & \vdots \\ s(N-1, 0, u, v) & s(N-1, 1, u, v) & \cdots & s(N-1, N-1, u, v) \end{bmatrix} \end{matrix}$$

Matrix-based transforms – 2D using basis images

Inverse transform

Each $S_{u,v}$ is a basis image

$$S_{u,v} = \begin{matrix} \begin{matrix} x \downarrow \\ y \rightarrow \end{matrix} \end{matrix} \begin{bmatrix} s(0, 0, u, v) & s(0, 1, u, v) & \cdots & s(0, N-1, u, v) \\ s(1, 0, u, v) & s(1, 1, u, v) & \cdots & s(1, N-1, u, v) \\ \vdots & \vdots & \ddots & \vdots \\ s(N-1, 0, u, v) & s(N-1, 1, u, v) & \cdots & s(N-1, N-1, u, v) \end{bmatrix}$$

If $s(x, y, u, v)$ is separable and symmetric, then

$$S_{u,v} = \mathbf{s}_u \mathbf{s}_v^H \text{ for complex vectors}$$

$$S_{u,v} = \mathbf{s}_u \mathbf{s}_v^T \text{ for real vectors}$$

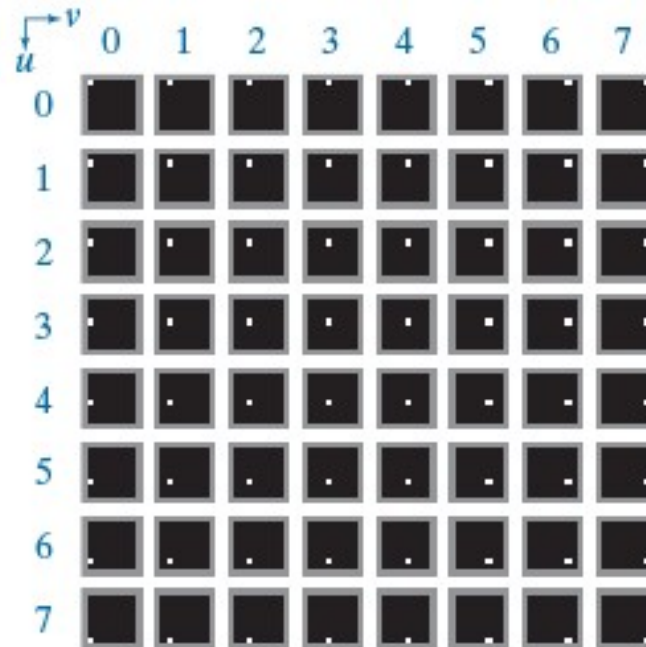
where

$$\begin{matrix} x \downarrow \\ \mathbf{s}_u = \end{matrix} \begin{bmatrix} s(0, u) \\ s(1, u) \\ \vdots \\ s(N-1, u) \end{bmatrix} \text{ and } \begin{matrix} y \downarrow \\ \mathbf{s}_v = \end{matrix} \begin{bmatrix} s(0, v) \\ s(1, v) \\ \vdots \\ s(N-1, v) \end{bmatrix}$$

Each basis image is an outer product of two basis vectors

Basis images of matrix-based 2D transforms

Standard basis images (for reference)

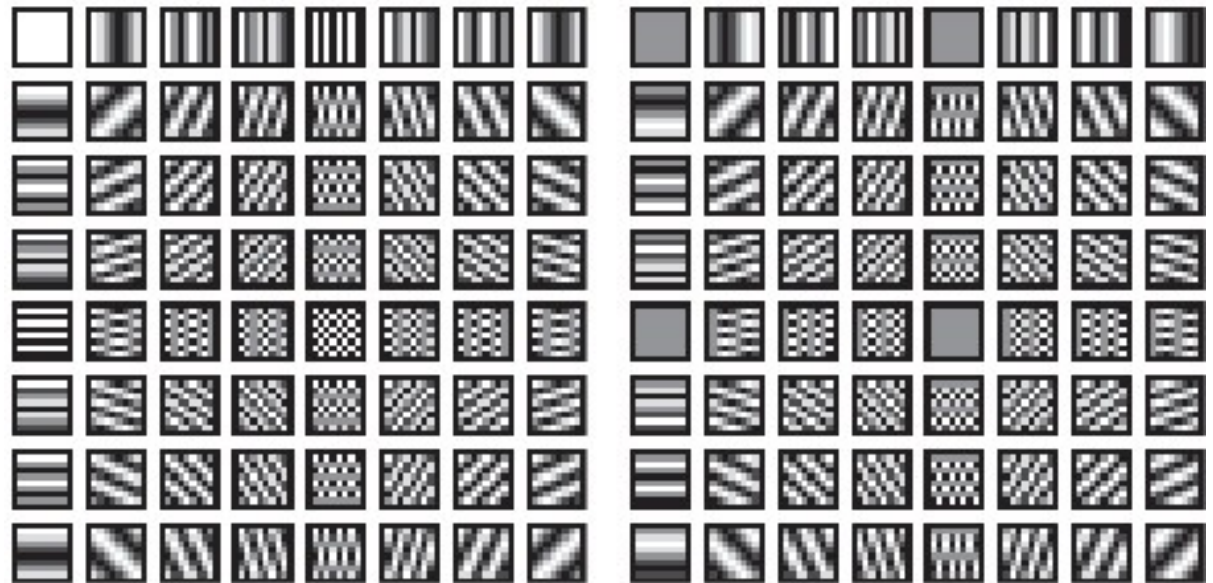


8-by-8
array of
8-by-8
basis images

Basis images of matrix-based 2D transforms

Discrete Fourier transform (DFT) basis images

$$\begin{aligned} X_k &= \sum_{n=0}^{N-1} x_n \cdot e^{-\frac{i2\pi}{N} kn} \\ &= \sum_{n=0}^{N-1} x_n \cdot \left[\cos\left(\frac{2\pi}{N} kn\right) - i \cdot \sin\left(\frac{2\pi}{N} kn\right) \right], \end{aligned}$$

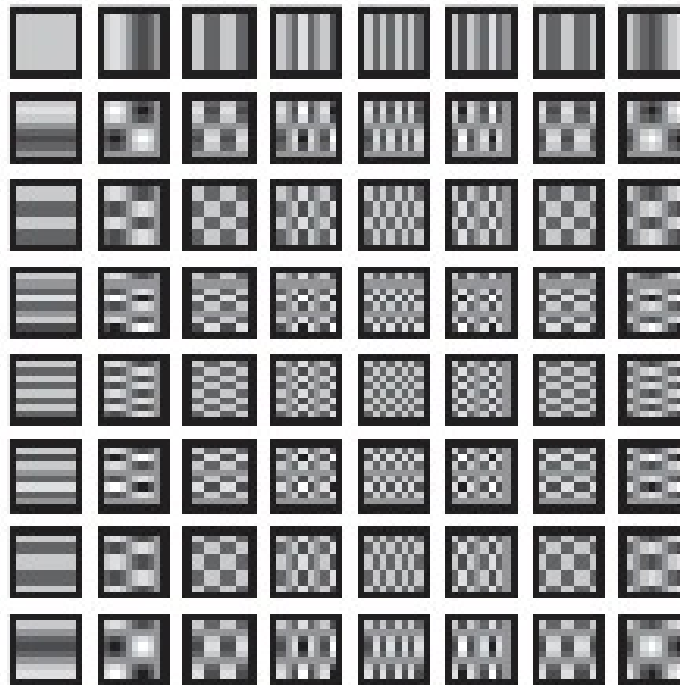


real part

imaginary part

Basis images of matrix-based 2D transforms

Discrete Hartley transform (DHT) basis images



$$H_k = \sum_{n=0}^{N-1} x_n \operatorname{cas}\left(\frac{2\pi}{N}nk\right) = \sum_{n=0}^{N-1} x_n \left[\cos\left(\frac{2\pi}{N}nk\right) + \sin\left(\frac{2\pi}{N}nk\right) \right] \quad k = 0, \dots, N-1.$$

$$\cos(z) + \sin(z) = \sqrt{2} \cos\left(z - \frac{\pi}{4}\right)$$

Basis images of matrix-based 2D transforms

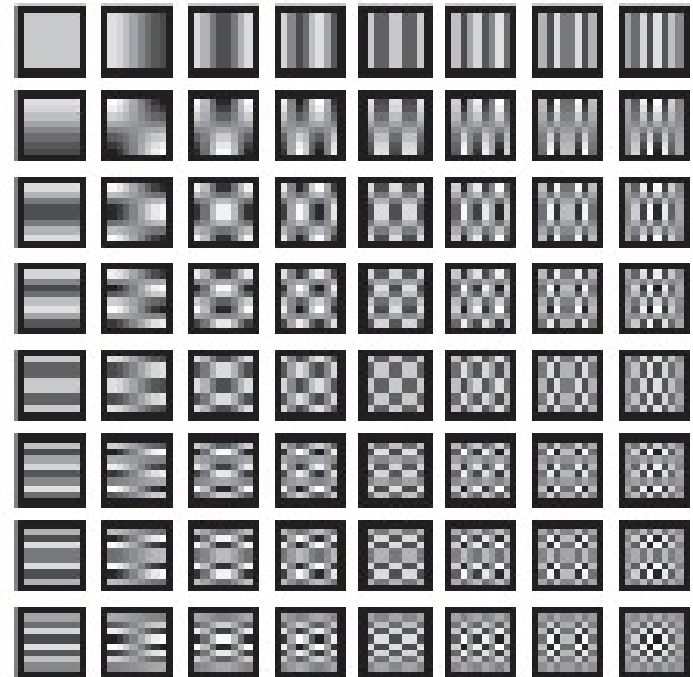
Discrete cosine transform (DCT) basis images

Inverse transformation kernel

$$s(x, u) = \alpha(u) \cos\left(\frac{(2x+1)u\pi}{2N}\right)$$

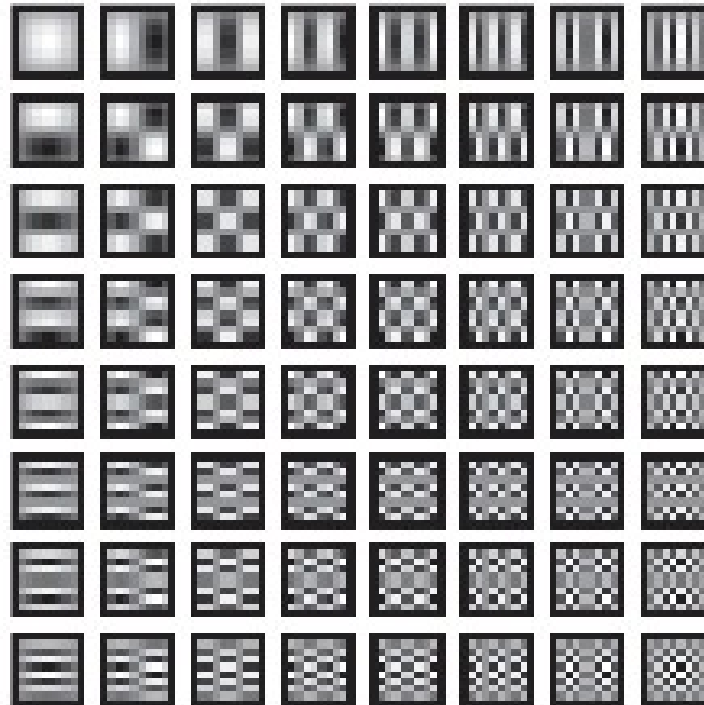
where

$$\alpha(u) = \begin{cases} \sqrt{\frac{1}{N}} & \text{for } u = 0 \\ \sqrt{\frac{2}{N}} & \text{otherwise} \end{cases}$$



Basis images of matrix-based 2D transforms

Discrete sine transform (DST) basis images



Basis images of matrix-based 2D transforms

Walsh-Hadamard transform (WHT) basis images

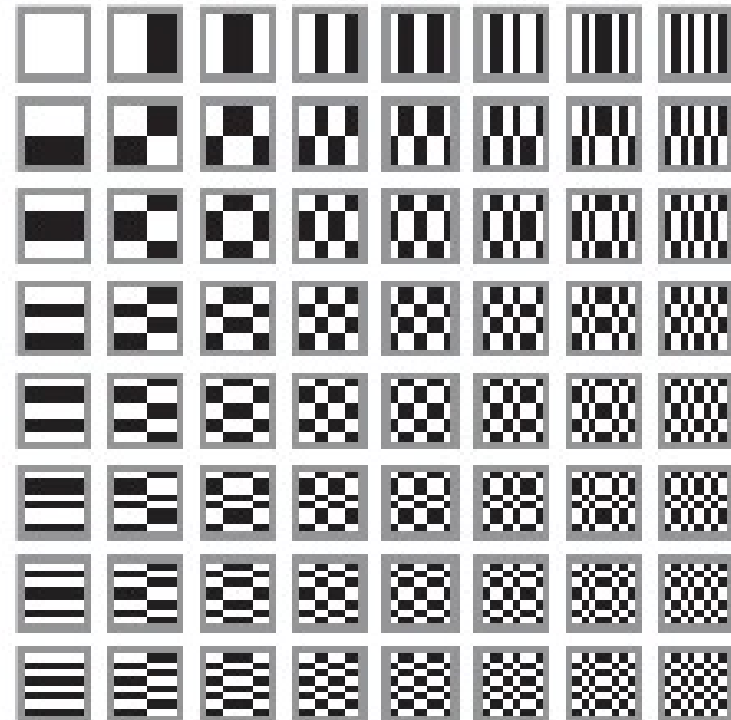
$$H_0 = +(1)$$

$$H_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$H_2 = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

$$H_3 = \frac{1}{2^{3/2}} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \end{pmatrix}$$

$$(H_n)_{i,j} = \frac{1}{2^{n/2}} (-1)^{i \cdot j}$$



Basis images of matrix-based 2D transforms

Haar transform (HAAR) basis images

Inverse transformation kernel

$$s(x, u) = \frac{1}{\sqrt{N}} h_u(x/N) \quad \text{for } x = 0, 1, \dots, N-1$$

where

$$h_u(x) = \begin{cases} 1 & \text{if } u = 0 \text{ and } 0 \leq x < 1 \\ 2^{p/2} & \text{if } u > 0 \text{ and } q/2^p \leq x < (q + 0.5)/2^p \\ -2^{p/2} & \text{if } u > 0 \text{ and } (q + 0.5)/2^p \leq x < (q + 1)/2^p \\ 0 & \text{otherwise} \end{cases}$$

where $p = \lceil \log_2 u \rceil$ and $q = 2^p - u$

